Willow Hughes

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About Me

Data Scientist passionate about AI safety and interpretability.

Education

MS **Western Washington University** — *Bellingham, WA*
Computer Science | 4.00 GPA — *Jan 2026 – Mar 2027*

BS **Western Washington University** — *Bellingham, WA*
Computer Science, Pre-Masters and Honors Curriculum | 3.84 GPA — *Jan 2023 – Dec 2025*

Experience

Research Intern — *Seattle, WA*
Pacific Northwest National Laboratory | National Security Directorate — *Jun 2026 - Present*

- Current intern researching AI Assurance

Deep Learning Research Assistant — *Bellingham, WA*
Hutchinson Machine Learning Research Group — *Dec 2025 – Present*

- Participating in a weekly reading group on foundational LLM research
- Developing segmentation models in PyTorch for LiDAR-based Maya site detection, reaching 0.74 mIoU on structures, collaborating with archaeologists to address domain-specific challenges
- Re-engineered the data pipeline on a fragile shared NFS/HPC-cluster (∼lossless 3TB to 70GB compression, staged caching, multiworker generation), cutting data generation from 24h to 3h and epoch time from 90m to 30m

Teaching Assistant, Object-Oriented Design — *Bellingham, WA*
Western Washington University — *Mar 2026 – June 2026*

Undergraduate Research Assistant — *Bellingham, WA*
Western Washington University — *Oct 2024 – Dec 2025*

- First author on research paper investigating Test-Driven Development in CS education; accepted to SIGCSE 2026 conference

Software Engineering Intern — *Seattle, WA*
Premera Blue Cross — *Jun 2024 – Aug 2024*

- Led the cloud migration of a C#/SQL web application to Azure, coordinating with Microsoft engineers to assist with strategy
- Built CI/CD (YAML) pipelines and IaC (Bicep) templates for core services, reducing deployment time by >40%
- Implemented authentication and authorization with Azure Entra ID

Projects

Conversational AI Spanish Tutor (Python, TypeScript, React, Flask, REST APIs)

- Building a real-time voice AI system with a Python/Flask REST API backend and React frontend, integrating STT, LLM, and TTS services
- Optimizing the end-to-end inference pipeline to achieve sub-second voice response latency

Multithreaded Image Processing Application (C, POSIX Threads)

- Developed a C program utilizing POSIX threads to implement a Laplacian edge detection algorithm on PPM images
- Engineered an efficient image filtering pipeline, achieving up to ∼70% runtime reduction through optimized thread synchronization and workload distribution

Skills

Languages: Python, Java, C, C#, SQL, TypeScript, JavaScript

Tools & Frameworks: Git, PyTorch, HuggingFace, NumPy, Pandas, Docker, Linux/Unix, Bash/Shell, CI/CD, Azure, AWS, GCP, REST APIs

Involvement

WWU's First Hackathon (Organizing Team) — *2026 – Present*

- Consulting with local nonprofits to source real-world projects for student teams to build during the event

Orchestra & Band Involvement — *2013 – Present*